Supplementary Material for "Dependable and Secure Privacy Trading in Federated Learning via MFG-RegretNet"

Kangkang Sun, Jianhua Li, *Senior Member, IEEE,* Xiuzhen Chen, *Member, IEEE,*
Weizhi Meng, *Senior Member, IEEE,* Minyi Guo, *Fellow, IEEE*

S1. Experimental Configuration and Additional Results

This section first gives the experimental configuration and then the complete results behind RQ1–RQ4 of the main manuscript, in the order of the research questions: the incentive results, with the attacks and the heterogeneous valuation scenarios, the cost of secure execution, the auction efficiency and the long-horizon privacy accounting, and the full confirmation table and convergence curves. All values use the budgets, valuation distributions, and seeds stated in the captions. The adaptive attacker, the cost of secure execution, the aggregation control, and the accounting curves under Gaussian RDP appear in Tables III, IV, and VII and Fig. 8 of the main manuscript.

A. Experimental Configuration

Datasets. MNIST [8] contains 70,000 grayscale handwritten-digit images (28×28 pixels, 10 classes), and CIFAR-10 [9] contains 60,000 RGB images (32×32 pixels, 10 classes). Both are partitioned among clients by a Dirichlet distribution $\mathrm{Dir}(\alpha)$ over class labels, whose draw fixes each client's per-class sample counts; α=0.5 gives mild and α=0.1 severe heterogeneity, standard settings in the FL literature [4].

MFG-RegretNet training. The encoder ϕ_{θ_h} maps the three features of Eq. (31) through two tanh layers of H=64 units with softplus-parameterized (non-negative) weights; the public network maps the budget feature $B/110$ and a feature of κ through two tanh layers of 64 units to $\mathbf{a}(B,\kappa) \in \mathbb{R}^{64}_{\geq 0}$ and $c(B,\kappa)$, and $S_{\max}$=50. Training uses Adam for 12,000 steps (learning rate 10^{-3}, then 3×10^{-5} for the last 2,000), gradient clipping at 1, and seed 42, on markets with $v_i \sim \mathrm{Uniform}[0,1]$, $\epsilon_i \sim \mathrm{Uniform}[0.1,5]$, $d_i = 10 + 490u_i^2$ with $u_i \sim \mathrm{Uniform}[0,1]$, budgets drawn half from $\mathrm{Uniform}[1,110]$ and half from $\mathrm{Uniform}[1,25]$, and κ log-uniform on $[0.003, 300]$. Each step uses a batch of 64 markets in double precision. The threshold table of Module 2 (Sec. VII-A) is built before training on 4,096 training profiles: for κ in $\{0.003, 0.01, 0.03, 0.1, 0.3, 1, 3, 10, 30, 100, 300\}$ and B in $\{1, 2, 5, 10, 20, 30, 50, 100, 110\}$, it keeps the threshold in $\{0, 0.005, \ldots, 0.25\}$ whose selection minimizes the mean, over the profiles, of the smallest J_κ that an aligned split among the selected clients attains on 400 purchase levels. It is interpolated bilinearly in $(\log\kappa, \log B)$, and a second table is built in the same way for fixed purchases. For the budgets of RQ1 and RQ3, $\tau = 0.015$ at κ=3 and $\tau = 0.02$ at κ=30.

TABLE S1
Privacy Valuation Distributions and Budget Settings

Scenario	Valuations and privacy budgets	Budget B
Uniform	$v_i \sim \mathrm{Uniform}[0,1]$, $\epsilon_i \sim \mathrm{Uniform}[0.1,5]$	25, 50, 75, 100
Bimodal	50% privacy-sensitive ($\epsilon_i \in [0.1, 0.5]$), 50% privacy-relaxed ($\epsilon_i \in [2,5]$)	50, 100
Realistic	$v_i \sim$ LogNormal(μ=0, σ=0.5), divided by its 99th percentile, capped at 1	100

Development and confirmation. The design of MFG-RegretNet (objective, selection threshold, and κ) was fixed in development runs on separate seeds; earlier variants of the design had also been evaluated on the official test sets with other seeds before it was fixed. The confirmation of RQ4 uses seeds 231–233, which no development run used, and a decision rule fixed before any of its runs.

Training of the regret-trained baselines. RegretNet and DM-RegretNet are trained for 200 epochs over 10,240 profiles in mini-batches of L=64 with SGD (learning rate 10^{-3}) and budgets drawn from $\mathrm{Uniform}[20, 110]$, so one network serves every budget of RQ1 and RQ3. Misreports start at the best point of a 5×9×3 grid over $\mathcal{B}_i$ and take R=25 PGA steps with η=0.01; multipliers are updated every 25 mini-batches with a dual step starting at 1 (regret) and 10 (IR) and growing by 1.5 per epoch. Both keep their original architectures (70,131 parameters, no b_{MFG}, a Softmax payment head that splits the budget) with input scaling; DM-RegretNet uses the error objective of [7] and a deterministic allocation rounded over the item and the abstention option. DM-RegretNet's training diverges in all five seeds between epochs 153 and 166: a single update inflates its weights by orders of magnitude, after which up to 90% of the clients are paid less than their privacy cost and every gradient is non-finite, so the weights no longer change. We therefore report its epoch-150 checkpoints, the last before any seed diverges; RegretNet is taken at epoch 200. Table S1 lists the valuation scenarios and Table S2 the FL settings of RQ4.

Baselines. Table S3 gives the reports, the release and

TABLE S2
FEDERATED LEARNING TRAINING SETTINGS OF RQ4 (N=10; BUDGETS AND PURCHASED PRIVACY AS IN TABLE VI)

Parameter	Value
Local epochs	$E = 1$
Batch (Poisson)	64
DP release	DP-SGD at $(\epsilon_i^{\text{out}}, \delta)$
Per-example clipping	1.0
Learning rate	0.1 / 0.05 (MNIST / CIFAR)
Total FL rounds	$T_{\text{FL}} = 100$
Privacy parameter	$\delta = 10^{-5}$
Aggregation	Weighted by ϵ_i^{out}

payment rule, and the adaptation of every baseline, as implemented in the code that produced the results.

B. Adaptation of Lin et al.

Lin et al. [5] enter the confirmation of RQ4 under the common protocol of all mechanisms: they share the partition, initial CNN, noise streams, and Poisson DP-SGD (clipping 1, record-level adjacency, δ=10^{-5}, RDP accounting), and each mechanism's per-round budgets calibrate the DP-SGD noise. Their Algorithm 2 keeps its monotone discrete menu, recursive payments, and ironing: the menu is solved once for the ten types $\{0.1, 0.2, \ldots, 1\}$ with equal weights, payments bind the IR constraint of the highest type and the adjacent upward IC constraints, and the menu is scaled so that its largest release is 5 and ten largest payments fit the budget B. Each client takes the contract with the highest utility among those within its declared budget, or exits. Replacing its local perturbation by the common DP-SGD makes it an adaptation, not a native reproduction.

C. Incentive Compatibility (RQ1)

Table S4 gives the RQ1 measurements of MFG-RegretNet at both budgets, and Table S5 compares the two adaptations of CSRA.

D. Attacks (RQ1)

Table S6 gives the coalition gains of Fig. 6 of the main manuscript per attack mode.

E. Heterogeneous Valuations (RQ1)

Table S7 gives the results in the heterogeneous valuation scenarios of Table S1. The size gains come from clients whose data share lies just above τ: a slightly smaller size takes them out of the selection and raises their dividend by more than their surplus from the release; without the dividend, no size misreport gains.

F. Cost of Secure Execution (RQ2)

Table IV of the main manuscript gives the measurements of RQ2: the secure round of MFG-RegretNet with secret-shared valuations and budgets and public weights and verified data sizes, the oblivious sort that PAC, VCG, and CSRA need (a lower bound on their secure cost), and the TEE channel cost (bid decryption plus outcome signing). The SMPC columns are measured on one server (Intel Xeon Silver 4316, four threads per party), the TEE column on the workstation of the earlier measurement. The secure round, with the selection, the projection, the payment layer, and the dividend, is measured directly at every N on one server; a cost model of measured primitive costs times operation counts predicts its rounds exactly, its traffic within 0.3%, and its time within 25%. With CrypTen's default fixed-point approximations, the secure releases and payments deviate from the plaintext round by at most 0.027. At $N \geq 200$ the selection keeps a single client (limitation L7 of the main manuscript), so the release equals that client's budget; the round still prices every client.

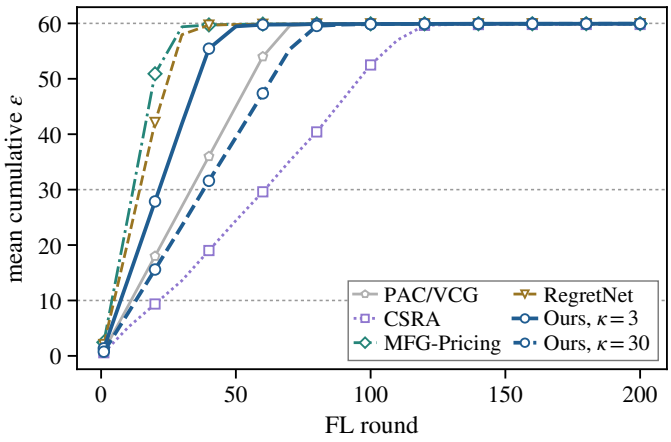


Fig. S1. Laplace cumulative privacy accounting over 200 rounds (B=50, three seeds), with the pre-release check at E_i^{tot}=60.

G. Auction Efficiency (RQ3)

Table S8 splits the welfare of Table V of the main manuscript into its parts.

H. Long-Horizon Privacy Accounting (RQ3)

Fig. S1 shows the mean cumulative privacy under Laplace composition, as Fig. 8 of the main manuscript does under Gaussian RDP, and Tables S9 and S10 give, for both, the rounds at which the first client reaches each level.

I. Confirmation Results and Convergence (RQ4)

Table S11 lists the final accuracy of every mechanism in the 18 settings behind Table VI of the main manuscript, Table S12 the clients' cumulative privacy after the 100 rounds, and Figs. S2 and S3 the test accuracy per round in every setting.

TABLE S3

BASELINES AS IMPLEMENTED UNDER THE COMMON PROTOCOL: THE REPORTS EACH RULE READS, ITS RELEASE AND PAYMENT, AND HOW IT IS ADAPTED. ALL RUN ONE AUCTION PER ROUND ON THE SAME REPORTS; A RELEASE ABOVE A CLIENT'S TRUE BUDGET IS NEITHER EXECUTED NOR PAID AT SETTLEMENT; RQ4 PASSES EVERY RELEASE TO THE SAME DP-SGD EXECUTOR. $v_{(j)}$ IS THE j-TH SMALLEST REPORTED VALUATION.

Mechanism	Reads	Release and payment	Adaptation and use
PAC [1]	$\hat{v}$	Winners: the largest k with $v_{(k)}/(N-k) \le B/k$; release $1/(N-k)$ each; payment $\min\{B/k,\, v_{(k+1)}/(N-k)\}$.	The selection and payment rule of FairQuery, adopted for this market (Algorithm 1 of the main manuscript). RQ1–RQ4.
VCG [2]	$\hat{v}$	Winners: the largest k with $k\, v_{(k+1)}/(N-k) \le B$; release $1/(N-k)$ each; payment $v_{(k+1)}/(N-k)$.	Reverse procurement with threshold payments, restricted to budget-feasible k. RQ1–RQ4.
PAC, VCG (capped)	$\hat{v}, \hat{\epsilon}$	As PAC or VCG, with each release lowered to $\min\{1/(N-k), \hat{\epsilon}_i\}$ and the payment unchanged.	Comparators kept from earlier versions; no incentive claim for the reported budget. RQ4.
CSRA [3]	$\hat{v}, \hat{\epsilon}, \hat{d}$	Quality $q_i = (\hat{\epsilon}_i/\sum_j \hat{\epsilon}_j)(\hat{d}_i/\sum_j \hat{d}_j)$, bid $b_i = \hat{v}_i\hat{\epsilon}_i$; winners: the first k clients by decreasing q_i/b_i, k the largest with $(b_{k+1}/q_{k+1})\sum_{i\le k} q_i \le B$; payment $(b_{k+1}/q_{k+1})\, q_i$; release $1/(N-k)$ or $\hat{\epsilon}_i$.	One round, without reputation history or anomaly detection. Release $1/(N-k)$ in RQ1–RQ4; release $\hat{\epsilon}_i$ in RQ1 and RQ3 (Table S5).
MFG-Pricing [4]	$\hat{\epsilon}$	Every client releases $\hat{\epsilon}_i$; payment $B\hat{\epsilon}_i/\sum_j \hat{\epsilon}_j$.	No auction; the payment ignores valuations, so IR is not guaranteed. RQ1–RQ4.
Posted price	$\hat{v}, \hat{\epsilon}$	Public price $\pi(B)$, interpolated through $(5, 0.44)$, $(10, 0.63)$, $(15, 0.77)$, $(20, 0.89)$, $(25, 1)$, $(110, 1)$; a client with $\hat{v}_i \le \pi$ releases $\hat{\epsilon}_i \min\{1, B/(\pi\sum_j \hat{\epsilon}_j)\}$, the sum over accepting clients; payment π per unit released.	Price schedule fixed in advance. RQ4.
Lin et al. [5]	$\hat{v}, \hat{\epsilon}$	The menu of their Algorithm 2 on ten types $\{0.1, 0.2, \ldots, 1\}$ with equal weights, scaled so that the largest release is 5 and ten largest payments fit B; each client takes the contract with the highest utility among those within $\hat{\epsilon}_i$, or exits.	The common DP-SGD replaces their local perturbation (Sec. S1.B). RQ4.
RegretNet [6]	$\hat{v}, \hat{\epsilon}, \hat{d}$	Learned allocation and payment networks with the input scaling of MFG-RegretNet and without b_{MFG}.	RQ1–RQ3: trained to maximize the purchased privacy weighted by data size, with Lagrangian regret and IR penalties (seeds 42–46, 200 epochs). RQ4: the same architecture retrained to maximize the purchase weighted by data share, with its mean searched regret constrained to at most $0.02\, B/N$ (seed 42).
DM-RegretNet [7]	$\hat{v}, \hat{\epsilon}, \hat{d}$	Learned networks with deterministic allocation over $\{\text{item}, \text{abstain}\}$ and the same input scaling.	Trained to minimize the FL error bound with the same penalties; the epoch-150 checkpoint, before its training diverges. RQ1–RQ4.

TABLE S4

RQ1 DETAIL FOR MFG-REGRETNET (10,000 HELD-OUT PROFILES, UTILITIES WITH THE DIVIDEND): REGRET OF THE FULL SEARCH (VALUATION, BUDGET, AND DATA SIZE, WITH THE THRESHOLD CROSSINGS), WITH VERIFIED DATA SIZES, AND FROM SIZE MISREPORTS ALONE; SHARE OF CONTRACTS WITH NEGATIVE UTILITY; BUDGET USE; MARKETS WITHOUT A PURCHASE; PAYMENT PER UNIT OF RELEASED PRIVACY (MEAN, AND SHARE PAID AT THE PRICE CAP $\bar{v}$=1).

		Regret						No	Payment per unit	
B	κ	mean	max	verified, max	size only, max	IR viol.	$\mathcal{R}/B$	purchase	mean	at cap
25	3	5.4×10^{-9}	3.3×10^{-6}	3.2×10^{-6}	$< 10^{-12}$	0	0.956	0	0.995	48%
25	30	1.9×10^{-9}	2.4×10^{-6}	2.4×10^{-6}	$< 10^{-12}$	0	0.968	0	0.993	33%
50	3	5.6×10^{-9}	3.2×10^{-6}	3.1×10^{-6}	$< 10^{-12}$	0	0.978	0	0.995	48%
50	30	2.1×10^{-9}	2.4×10^{-6}	2.4×10^{-6}	$< 10^{-12}$	0	0.984	0	0.993	33%

TABLE S5

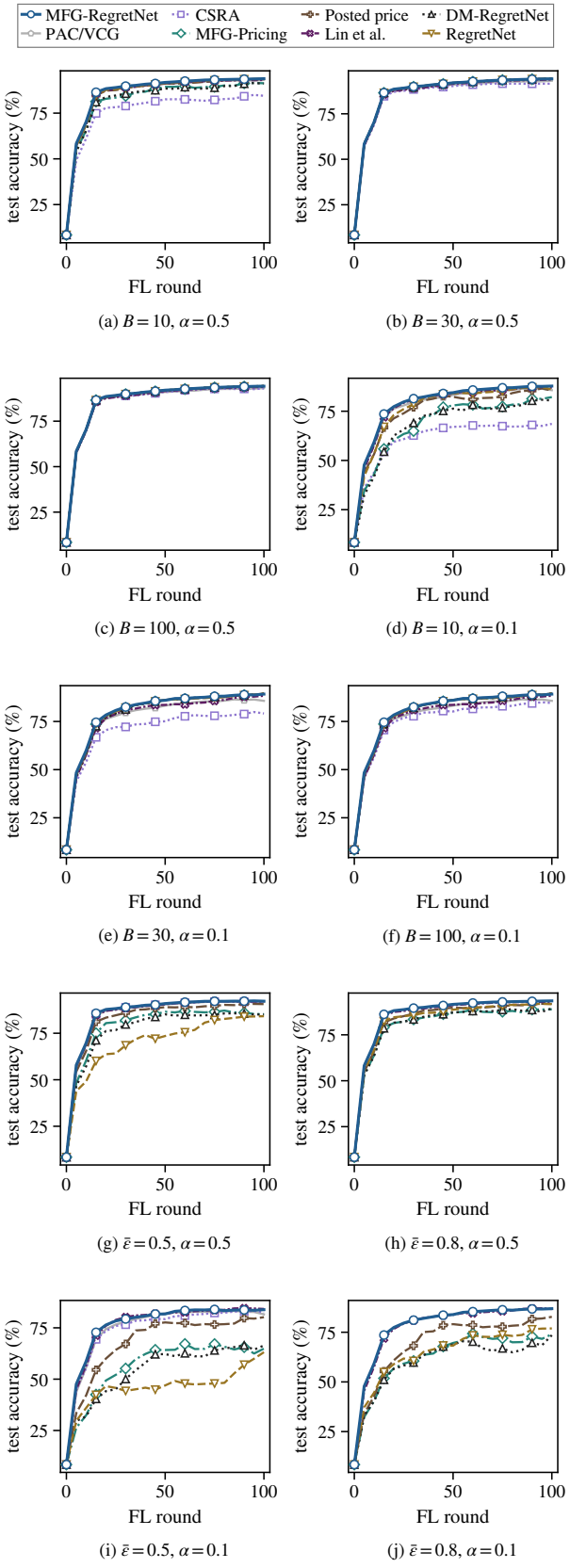
CSRA WITH RELEASES EQUAL TO THE DECLARED BUDGETS, AGAINST THE RELEASE-$1/(N-k)$ ADAPTATION USED IN RQ4 (10,000 PROFILES); WINNERS, PAYMENTS, AND REVENUE ARE THE SAME UNDER BOTH RULES.

	Release $1/(N-k)$		Declared release	
	B=25	B=50	B=25	B=50
Regret (10^{-3})	261	221	71.0	124
IR viol.	1.5%	2.3%	0	0
$\mathcal{R}/B$	0.734	0.638	0.734	0.638
$\sum_i \epsilon_i^{\text{out}}$	2.87	5.15	17.54	20.34
	Budget B			
	25	50	75	100
$\overline{\text{SW}}$				
release $1/(N-k)$	17.14	29.56	39.56	47.50
declared release	10.98	22.65	32.60	40.60
Privacy cost $\sum_i v_i \epsilon_i^{\text{out}}$				
release $1/(N-k)$	1.21	2.34	2.99	3.36
declared release	7.36	9.25	9.94	10.26
Revenue $\mathcal{R}$	18.34	31.90	42.55	50.87

TABLE S6

COALITION GAIN UNDER EXTREME COORDINATED ATTACKS (% OF THE TRUTHFUL UTILITY; B=50, SEEDS 42–44, COALITION RATIOS 0.1–0.5 POOLED; CSRA: RELEASE $1/(N-k)$).

Mechanism	OB	UB	ϵ^+	ϵ^-	UB+ϵ^-	UB+ϵ^+
PAC/VCG	−18.4	−14.1	0.0	0.0	−14.1	−14.1
CSRA	+2.0	−111.4	+34.9	−79.6	−111.4	−111.4
MFG-Pricing	0.0	0.0	+18.1	−65.3	−65.3	+18.1
RegretNet	+0.3	−0.6	+11.0	−23.0	−23.6	+10.3
MFG-RegretNet						
κ=3	+0.7	−1.0	+0.6	−5.7	−7.3	−0.3
κ=30	+0.5	−0.8	−0.5	−0.6	−1.5	−1.2



(a) $B=10$, $\alpha=0.5$ (b) $B=30$, $\alpha=0.5$ (c) $B=100$, $\alpha=0.5$ (d) $B=10$, $\alpha=0.1$ (e) $B=30$, $\alpha=0.1$ (f) $B=100$, $\alpha=0.1$ (g) $\bar{\varepsilon}=0.5$, $\alpha=0.5$ (h) $\bar{\varepsilon}=0.8$, $\alpha=0.5$ (i) $\bar{\varepsilon}=0.5$, $\alpha=0.1$ (j) $\bar{\varepsilon}=0.8$, $\alpha=0.1$

Fig. S2. Test accuracy per FL round on MNIST in every setting of the confirmation, mean over seeds 231–233, smoothed as in Fig. 9 of the main manuscript; a mechanism that no budget brings to $\bar{\varepsilon}$ in every seed has no curve.

TABLE S7

HETEROGENEOUS VALUATION SCENARIOS OF TABLE S1, NETWORK TRAINED ON THE UNIFORM SCENARIO (5,000 PROFILES EACH, UTILITIES WITH THE DIVIDEND): REGRET OF THE FULL SEARCH, WITH VERIFIED DATA SIZES, AND FROM SIZE MISREPORTS ALONE; CLIENTS (OF 50,000) WHOSE SIZE MISREPORT GAINS; SHARE OF CONTRACTS WITH NEGATIVE UTILITY; BUDGET USE; MARKETS WITHOUT A PURCHASE; PAYMENT PER UNIT OF RELEASED PRIVACY.

		Regret, max			Size gain			No	Payment per unit	
Scenario, B	κ	full	verified	size only	(clients)	IR viol.	$\mathcal{R}/B$	purchase	mean	at cap
Bimodal, 50	3	5.7×10^{-2}	8.3×10^{-6}	5.7×10^{-2}	2	0	0.975	0	0.997	66%
Bimodal, 50	30	1.3×10^{-6}	1.3×10^{-6}	$< 10^{-12}$	0	0	0.983	0	0.995	58%
Bimodal, 100	3	5.5×10^{-2}	6.1×10^{-6}	5.5×10^{-2}	1	0	0.988	0	0.997	66%
Bimodal, 100	30	9.9×10^{-7}	9.9×10^{-7}	$< 10^{-12}$	0	0	0.992	0	0.996	59%
Realistic, 100	3	4.3×10^{-6}	4.3×10^{-6}	$< 10^{-12}$	0	0	0.965	0	0.993	38%
Realistic, 100	30	1.8×10^{-6}	1.8×10^{-6}	$< 10^{-12}$	0	0	0.974	0	0.991	19%

TABLE S8

RQ3 WELFARE DECOMPOSITION PER ROUND (TRUTHFUL REPORTS, 10,000 PROFILES PER BUDGET): TOTAL PAYMENT $\mathcal{R}$, PAYMENT FOR THE RELEASE, DIVIDEND, PRIVACY COST $\sum_i v_i\epsilon_i^{\text{out}}$ UNDER THE TRUE VALUATIONS, CLIENT SURPLUS $\overline{\text{SW}}$ WITHOUT AND WITH THE DIVIDEND, RELEASED PRIVACY, AND SHARE OF MARKETS WITHOUT A PURCHASE. THE LEARNED BASELINES APPEAR IN TABLE V OF THE MAIN MANUSCRIPT.

B	Mechanism	$\mathcal{R}$	Release	Dividend	Privacy cost	$\overline{\text{SW}}$ w/o div.	$\overline{\text{SW}}$	$\sum_i \epsilon_i^{\text{out}}$	No purchase
25	MFG-RegretNet, κ=3	23.91	13.91	10.00	6.98	6.93	16.93	13.98	0
	MFG-RegretNet, κ=30	24.19	7.81	16.38	3.92	3.89	20.27	7.87	0
	PAC / VCG	8.18	8.18	–	4.10	4.08	4.08	9.00	0
	CSRA, release $1/(N-k)$	18.34	18.34	–	1.21	17.14	17.14	2.87	0
	CSRA, declared release	18.34	18.34	–	7.36	10.98	10.98	17.54	0
	MFG-Pricing	25.00	25.00	–	12.77	12.23	12.23	25.47	0
50	MFG-RegretNet, κ=3	48.91	13.99	34.92	7.02	6.97	41.89	14.06	0
	MFG-RegretNet, κ=30	49.19	7.83	41.36	3.93	3.90	45.26	7.88	0
	PAC / VCG	8.18	8.18	–	4.10	4.08	4.08	9.00	0
	CSRA, release $1/(N-k)$	31.90	31.90	–	2.34	29.56	29.56	5.15	0
	CSRA, declared release	31.90	31.90	–	9.25	22.65	22.65	20.34	0
	MFG-Pricing	50.00	50.00	–	12.77	37.23	37.23	25.47	0
75	MFG-RegretNet, κ=3	73.93	13.97	59.96	7.01	6.96	66.92	14.04	0
	MFG-RegretNet, κ=30	74.20	7.80	66.40	3.92	3.89	70.28	7.86	0
	PAC / VCG	8.18	8.18	–	4.10	4.08	4.08	9.00	0
	CSRA, release $1/(N-k)$	42.55	42.55	–	2.99	39.56	39.56	6.42	0
	CSRA, declared release	42.55	42.55	–	9.94	32.60	32.60	21.36	0
	MFG-Pricing	75.00	75.00	–	12.77	62.23	62.23	25.47	0
100	MFG-RegretNet, κ=3	98.96	13.88	85.08	6.97	6.91	91.99	13.95	0
	MFG-RegretNet, κ=30	99.22	7.76	91.47	3.89	3.86	95.33	7.81	0
	PAC / VCG	8.18	8.18	–	4.10	4.08	4.08	9.00	0
	CSRA, release $1/(N-k)$	50.87	50.87	–	3.36	47.50	47.50	7.14	0
	CSRA, declared release	50.87	50.87	–	10.26	40.60	40.60	21.84	0
	MFG-Pricing	100.00	100.00	–	12.77	87.23	87.23	25.47	0

TABLE S9

CUMULATIVE PRIVACY ACCOUNTING UNDER GAUSSIAN RDP (B=50, SEEDS 42–44, 200 ROUNDS): ROUND AT WHICH THE FIRST CLIENT REACHES ϵ^{tot}=10, 30, AND 60 (RANGE OVER SEEDS; –: NOT WITHIN 200 ROUNDS), AND RELEASES REFUSED BY THE PRE-RELEASE CHECK AT ϵ^{tot}=60 (MEAN OVER SEEDS).

Mechanism	ϵ^{tot}=10	30	60	Refused
PAC / VCG	75–77	–	–	0
CSRA	158–191	–	–	0
MFG-Pricing	4–7	36–41	–	619
RegretNet	6–10	57–64	–	64
MFG-RegretNet, κ=3	19–22	140–151	–	0
MFG-RegretNet, κ=30	67–76	–	–	0

TABLE S10

CUMULATIVE PRIVACY ACCOUNTING UNDER LAPLACE COMPOSITION, AS IN TABLE S9.

Mechanism	ϵ^{tot}=10	30	60	Refused
PAC / VCG	10	30–31	62–64	0
CSRA	18–19	58–59	–	761
MFG-Pricing	3–4	9–11	–	1743
RegretNet	3–4	12–13	–	1689
MFG-RegretNet, κ=3	5–7	17–21	–	1477
MFG-RegretNet, κ=30	9–11	28–34	–	1147

TABLE S11

RQ4 CONFIRMATION: FINAL TEST ACCURACY $\mathcal{A}_{\text{final}}$ (%), MEAN OVER SEEDS 231–233, OF MFG-REGRETNET AND THE EXTERNAL MECHANISMS IN ALL 18 SETTINGS; BOLD MARKS THE BEST EXTERNAL MECHANISM OF EACH SETTING, AND – MARKS A MECHANISM THAT NO BUDGET IN $[0.5, 110]$ BRINGS TO THE PURCHASED PRIVACY $\bar{\epsilon}$ IN EVERY SEED.

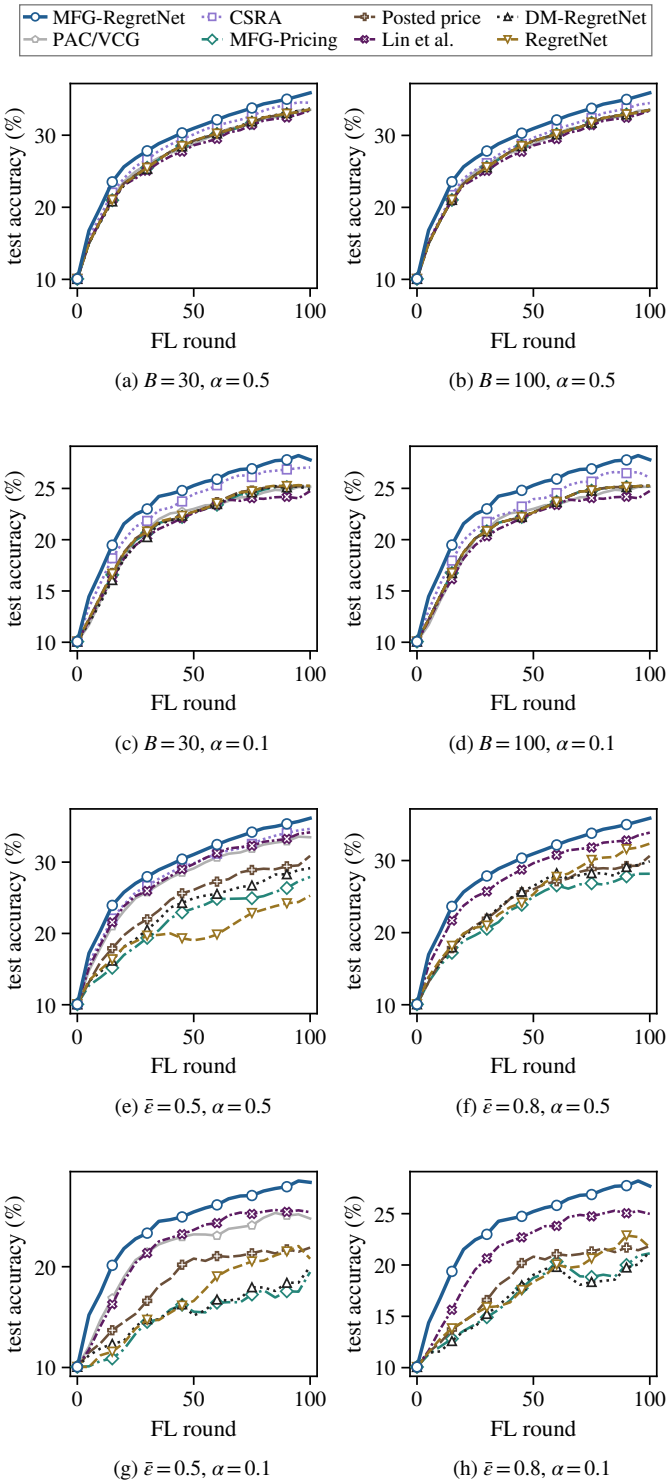
	α=0.5					α=0.1				
MNIST	B=10	B=30	B=100	$\bar{\epsilon}$=0.5	$\bar{\epsilon}$=0.8	B=10	B=30	B=100	$\bar{\epsilon}$=0.5	$\bar{\epsilon}$=0.8
MFG-RegretNet (Ours)	93.8	94.2	94.2	92.2	93.5	87.8	89.2	89.2	83.8	86.9
PAC	93.2	93.2	93.2	91.9	–	85.7	85.7	85.7	81.6	–
PAC (capped)	93.3	93.3	93.3	91.9	**93.3**	86.7	86.7	86.7	82.7	86.5
VCG	93.2	93.2	93.2	**92.1**	–	85.7	85.7	85.7	81.2	–
VCG (capped)	93.3	93.3	93.3	91.9	93.2	86.7	86.7	86.7	82.6	86.6
CSRA	84.5	91.5	93.0	91.7	–	68.6	79.2	84.6	82.5	–
MFG-Pricing	91.3	94.3	**94.3**	85.3	88.9	82.2	**89.5**	89.5	64.9	74.0
Posted price	93.6	94.3	94.3	90.7	91.8	85.5	89.5	89.5	80.2	82.8
Lin et al.	93.1	93.9	93.9	92.1	93.0	**86.9**	88.3	88.3	**84.2**	**86.7**
DM-RegretNet	91.1	94.3	94.3	85.1	89.3	80.4	89.5	89.5	66.4	73.5
RegretNet	**93.6**	**94.3**	94.3	84.1	91.7	86.1	89.5	**89.5**	63.1	76.9

	α=0.5				α=0.1			
CIFAR-10	B=30	B=100	$\bar{\epsilon}$=0.5	$\bar{\epsilon}$=0.8	B=30	B=100	$\bar{\epsilon}$=0.5	$\bar{\epsilon}$=0.8
MFG-RegretNet (Ours)	35.9	35.9	36.2	35.9	27.8	27.8	28.4	27.7
PAC	33.5	33.5	33.5	–	25.2	25.2	24.8	–
PAC (capped)	34.0	34.0	33.9	34.0	25.7	25.7	25.2	25.6
VCG	33.5	33.5	33.6	–	25.2	25.2	24.8	–
VCG (capped)	34.0	34.0	33.9	**34.0**	25.7	25.7	25.3	**25.7**
CSRA	**34.5**	**34.5**	**34.7**	–	**27.0**	**26.0**	–	–
MFG-Pricing	33.5	33.5	27.9	28.2	25.2	25.2	19.5	21.1
Posted price	33.5	33.5	30.9	30.6	25.2	25.2	21.9	21.8
Lin et al.	33.4	33.4	34.2	33.9	24.7	24.7	**25.4**	25.0
DM-RegretNet	33.7	33.5	29.1	29.9	24.9	25.2	20.0	21.3
RegretNet	33.6	33.5	25.3	32.3	25.3	25.3	20.8	21.7

TABLE S12

PER-CLIENT CUMULATIVE PRIVACY IN THE RQ4 CONFIRMATION: EACH CLIENT'S ϵ AFTER THE 100 ROUNDS (δ=10^{-5}, ADD/REMOVE ADJACENCY, RDP COMPOSITION OF ITS DP-SGD PARTICIPATIONS), AS THE MEAN, THE 95TH PERCENTILE, AND THE MAXIMUM OVER THE 30 CLIENT-SEED PAIRS OF A SETTING, AND THE MEAN NUMBER OF ROUNDS IN WHICH A CLIENT PARTICIPATES, FOR MFG-REGRETNET AND THE BEST EXTERNAL MECHANISM OF EACH SETTING (TABLE S11).

	MFG-RegretNet				Best external mechanism				
Setting	mean	P95	max	rounds	mechanism	mean	P95	max	rounds
MNIST									
α=0.5, B=10	6.2	7.3	7.5	100	RegretNet	10.8	12.2	12.2	92
α=0.5, B=30	9.0	10.4	10.8	100	RegretNet	15.3	17.5	18.7	100
α=0.5, B=100	8.9	10.4	10.6	100	MFG-Pricing	15.7	18.2	19.3	100
α=0.5, $\bar{\epsilon}$=0.5	3.5	4.6	4.8	83	VCG	4.1	5.1	5.7	74
α=0.5, $\bar{\epsilon}$=0.8	5.2	6.3	6.5	97	PAC (capped)	5.5	6.8	7.5	89
α=0.1, B=10	5.6	8.6	8.9	80	Lin et al.	6.1	8.8	8.9	89
α=0.1, B=30	8.6	10.7	10.9	97	MFG-Pricing	16.2	21.1	24.5	99
α=0.1, B=100	8.5	10.6	10.9	97	RegretNet	16.2	21.6	24.4	100
α=0.1, $\bar{\epsilon}$=0.5	3.2	4.9	5.0	77	Lin et al.	4.1	5.5	5.8	95
α=0.1, $\bar{\epsilon}$=0.8	4.7	7.1	7.2	80	Lin et al.	5.8	8.4	8.4	90
CIFAR-10									
α=0.5, B=30	5.8	7.1	7.3	100	CSRA	4.1	4.8	4.9	71
α=0.5, B=100	5.8	7.1	7.2	100	CSRA	5.5	6.0	6.2	74
α=0.5, $\bar{\epsilon}$=0.5	3.9	4.9	4.9	100	CSRA	4.3	4.7	4.8	72
α=0.5, $\bar{\epsilon}$=0.8	5.5	6.8	6.8	100	VCG (capped)	5.8	7.0	7.7	89
α=0.1, B=30	5.1	7.1	7.3	90	CSRA	2.9	3.7	3.7	68
α=0.1, B=100	5.0	7.0	7.2	90	CSRA	4.3	6.3	6.5	73
α=0.1, $\bar{\epsilon}$=0.5	3.7	5.0	5.3	93	Lin et al.	4.3	5.8	6.0	95
α=0.1, $\bar{\epsilon}$=0.8	5.3	7.0	7.4	97	VCG (capped)	6.0	8.4	9.2	89



(a) $B = 30$, $\alpha = 0.5$ (b) $B = 100$, $\alpha = 0.5$

(c) $B = 30$, $\alpha = 0.1$ (d) $B = 100$, $\alpha = 0.1$

(e) $\bar{\varepsilon} = 0.5$, $\alpha = 0.5$ (f) $\bar{\varepsilon} = 0.8$, $\alpha = 0.5$

(g) $\bar{\varepsilon} = 0.5$, $\alpha = 0.1$ (h) $\bar{\varepsilon} = 0.8$, $\alpha = 0.1$

Fig. S3. Test accuracy per FL round on CIFAR-10 in every setting of the confirmation, as in Fig. S2.

S2. Formal Preliminaries

Table S13 summarizes the threat model of Sec. IV-B in the main manuscript: what each party sees, the attacks it may attempt, and the protections that apply.

Definition 1 (PLDP [10]) *For user i with $\epsilon_i{>}0, \delta_i{\geq}0$, mechanism $\mathcal{M}^i$ is (ϵ_i, δ_i)-PLDP if for all adjacent inputs $\boldsymbol{x}, \boldsymbol{x}'$ and measurable sets S of outputs:* $\Pr[\mathcal{M}^i(\boldsymbol{x}) \in S] \leq e^{\epsilon_i} \Pr[\mathcal{M}^i(\boldsymbol{x}') \in S] + \delta_i$.

FL global model: $\mathbf{w}_G{=}\sum_i \alpha_i \mathbf{w}_i$, $\mathbf{w}_G^*{=}\arg\min_{\mathbf{w}} \sum_i \alpha_i \mathcal{L}_i(\mathbf{w}_i)$. Gradient ℓ_2-sensitivity: $\Delta f_{\mathbf{w}_i} = \max_{\mathcal{D}_i \sim \mathcal{D}_i'} \|\nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i) - \nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i')\|_2$.

Sec. VIII-A1 of the main manuscript composes the per-round releases of a client as follows.

Proposition S1 (RDP Accounting for PAG-FL) *Let client i participate in the rounds $t \in \mathcal{H}_i$, and let its round-t release be $(\alpha, \rho_i^{(t)}(\alpha))$-RDP at every order $\alpha > 1$. Then its releases over all rounds are $(\alpha, \rho_i^{\mathrm{tot}}(\alpha))$-RDP with $\rho_i^{\mathrm{tot}}(\alpha) = \sum_{t \in \mathcal{H}_i} \rho_i^{(t)}(\alpha)$, and, for any $\delta' \in (0,1)$, they are $(\epsilon_i^{\mathrm{tot}}, \delta')$-DP with*

$$\epsilon_i^{\mathrm{tot}} \leq \min_{\alpha > 1} \Big\{ \rho_i^{\mathrm{tot}}(\alpha) + \frac{\log(1/\delta')}{\alpha - 1} \Big\}. \tag{S1}$$

For Gaussian releases with the classical calibration to $\epsilon_i^{(t),\mathrm{out}}$, this bound is Eq. (41) of the main manuscript.

S3. HJB-Fokker-Planck System

The mean-field bid statistic is $b_{\mathrm{MFG}}(t) = \mathbb{E}_{\mu_t}[b] = \int_{\mathcal{S}} b(s;\mu_t)\, d\mu_t(s)$. The value function $\Phi(s,t;\mu)$ satisfies the **HJB** equation:

$$-\partial_t \Phi - \mathcal{H}(s, \nabla_s \Phi, \mu_t) - \tfrac{\sigma^2}{2} \Delta_s \Phi = 0, \quad \Phi(s,T) = 0, \tag{S2}$$

with Hamiltonian $\mathcal{H} = \max_b [\tilde{p}(b;\mu_t) - c(v,\epsilon(b)) + (\nabla_s \Phi)^\top f(s,b,\mu_t)]$, the mean-field payment $\tilde{p}(b;\mu_t)$ of Assumption 4, and a Neumann condition at the boundary of $\mathcal{S}$, where the state is reflected (R6). The population density evolves via the **Fokker-Planck** equation:

$$\partial_t \mu_t + \nabla_s \cdot (\mu_t f(s, b^*, \mu_t)) = \tfrac{\sigma^2}{2} \Delta_s \mu_t, \quad \mu_0 = \mathcal{W}. \tag{S3}$$

Equations (S2) and (S3) form a coupled forward-backward system whose fixed point is the MFE [11].

S4. Proofs

This section proves the formal results of the main manuscript in the order in which they appear, followed by Proposition S1. Theorem, proposition, corollary, assumption, algorithm, and equation numbers without the prefix S refer to the main manuscript.

A. Proof of Theorem 1

Proof Sketch: Consider the threshold-based PAC mechanism (Algorithm 1), where valuations are sorted $v_1 \leq \cdots \leq v_n$, the largest k is selected such that $c(v_k, 1/(n{-}k)) \leq B/k$, and payments are $p_i = \min(B/k, c(v_{k+1}, 1/(n-k)))$ for $i \leq k$ and $p_i = 0$ otherwise.

TABLE S13
Threat model and protection status. HBC denotes honest-but-curious behavior; TEE, trusted execution environment; SMPC, secure multi-party computation.

Entity/Risk	Visible Information	Potential Attack	Protection Status
AB / FLB	Bids, winner set, payments	Bid inference, payment manipulation	DSIC handles misreporting; bid privacy requires TEE/SMPC
FL server	Aggregate update, global model	Gradient inversion from updates	DP noise; compatible secure aggregation
Clients	Own bids, public outcomes	False bidding, coordinated collusion	Single-client regret measured; coordinated stress tests in RQ1
External observer	Published outputs	Inferring losing bids or privacy preferences	Do not publish losing bids; release only aggregate statistics

DSIC (for $i \leq k$): Misreporting a higher valuation may cause i to drop out of the top-k, yielding $p_i' = 0$ and $u_i' = 0$. Since $u_i = p_i - c(v_i, \epsilon_i) \geq 0$ (by IR), deviation yields no gain.

DSIC (for $i > k$): Misreporting a lower valuation may enter the top-k, but the diluted payment $p_i' \leq B/(k+1)$ combined with the higher true cost results in $u_i' < 0$ or no improvement.

For the declared privacy budget, the allocation, the release $1/(n-k)$, and the payment of Algorithm 1 depend only on the reported valuations, so a unilateral report $\hat{\theta}_i = (\hat{v}_i, \hat{\epsilon}_i)$ yields the same utility as $(\hat{v}_i, \epsilon_i)$; in the terms of Proposition 1, the two cases above give condition (i) at every cap, and condition (ii) holds with equality. Hence truthful reporting of $\theta_i = (v_i, \epsilon_i)$ is a dominant strategy (DSIC), and all utilities are non-negative (IR).

BNE: By DSIC, for any fixed θ_{-i}, truthful reporting maximises $u_i(\theta_i, \cdot)$. Taking expectations over $\theta_{-i} \sim \mathcal{W}_{-i}$ yields Eq. (12), i.e., the BNE of Definition 3.

For finite N, existence follows from Nash's theorem. In the large-N regime, the multi-agent game converges to a Mean-Field Game, where fixed-point theorems from MFG literature [11] guarantee equilibrium existence. ∎

B. Proof of Proposition 1

Proof: Necessity. Fix a cap $\hat{\epsilon}$. Types $(v, \hat{\epsilon})$ may report any $(\hat{v}, \hat{\epsilon})$, so DSIC gives $\tilde{U}(v, \hat{\epsilon}) := p(v, \hat{\epsilon}) - v\, e(v, \hat{\epsilon}) = \max_{\hat{v}} \{p(\hat{v}, \hat{\epsilon}) - v\, e(\hat{v}, \hat{\epsilon})\}$. As a maximum of affine functions of v whose slopes $-e(\hat{v}, \hat{\epsilon}) \in [-\bar{\epsilon}, 0]$ are bounded, $\tilde{U}(\cdot, \hat{\epsilon})$ is convex and Lipschitz, its derivative is $-e(v, \hat{\epsilon})$ almost everywhere, and convexity makes $e(\cdot, \hat{\epsilon})$ non-increasing. Integrating gives $\tilde{U}(v, \hat{\epsilon}) = \tilde{U}(\bar{v}, \hat{\epsilon}) + \int_v^{\bar{v}} e(s, \hat{\epsilon})\, ds$, which is (i) with $h(\hat{\epsilon}) = \tilde{U}(\bar{v}, \hat{\epsilon})$ and $\tilde{U} = U$. A type (v, ϵ) with $\epsilon > \hat{\epsilon}$ may report $(v, \hat{\epsilon})$ and obtain $U(v, \hat{\epsilon})$, so DSIC gives $U(v, \epsilon) \geq U(v, \hat{\epsilon})$, which is (ii).

Sufficiency. For a type (v, ϵ) and a report $(\hat{v}, \hat{\epsilon})$ with $\hat{\epsilon} \leq \epsilon$, (i) gives $p(\hat{v}, \hat{\epsilon}) - v\, e(\hat{v}, \hat{\epsilon}) = U(\hat{v}, \hat{\epsilon}) - (v - \hat{v})\, e(\hat{v}, \hat{\epsilon})$, hence

$$U(v, \hat{\epsilon}) - \big[p(\hat{v}, \hat{\epsilon}) - v\, e(\hat{v}, \hat{\epsilon})\big] = \int_v^{\hat{v}} \big(e(s, \hat{\epsilon}) - e(\hat{v}, \hat{\epsilon})\big)\, ds \geq 0 \tag{S4}$$

by monotonicity, whichever of $v, \hat{v}$ is larger. By (ii), $U(v,\hat{\epsilon}) \le U(v,\epsilon)$, which is the truthful utility. ∎

For PAC, the release of a winner is $1/(n-k)$ and does not depend on $\hat{\epsilon}$, and its valuation-DSIC (Sec. IV-D) is condition (i) at every cap, so (ii) holds with equality. If p does not depend on $\hat{\epsilon}$ and $e = z(\hat{v})\hat{\epsilon}$, then $U(v,\hat{\epsilon}) = p(v) - v\,z(v)\hat{\epsilon}$ is strictly decreasing in $\hat{\epsilon}$ whenever $v\,z(v) > 0$, and the gain of declaring ϵ_{floor} is $v\,z(v)(\epsilon - \epsilon_{\text{floor}})$.

C. Proof of Lemma 1

Proof: By Eq. (17), $\mathbb{E}\|\tilde{\mathbf{w}}_G - \mathbf{w}_G^*\|_2^2 = \|\mathbf{w}_G - \mathbf{w}_G^*\|_2^2 + C_{\text{DP}}|\mathcal{I}|/S^2$ with $\mathbf{w}_G = \sum_i \alpha_i \mathbf{w}_i$. Since $\sum_i \alpha_i = \sum_i \pi_i = 1$, $\mathbf{w}_G - \mathbf{w}_G^* = \sum_i (\alpha_i - \pi_i)\mathbf{w}_i = \sum_i (\alpha_i - \pi_i)(\mathbf{w}_i - \mathbf{w}_G^*)$. The triangle and Cauchy–Schwarz inequalities give $\|\mathbf{w}_G - \mathbf{w}_G^*\|_2 \le \sum_i |\alpha_i - \pi_i|\,\|\mathbf{w}_i - \mathbf{w}_G^*\|_2 \le \|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2 \big(\sum_i \|\mathbf{w}_i - \mathbf{w}_G^*\|_2^2\big)^{1/2} \le \zeta\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2$ by Assumption 2. Squaring and adding the noise term gives $\zeta^2\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2^2 + C_{\text{DP}}|\mathcal{I}|/S^2 = C_{\text{DP}} J_\kappa$ with $\kappa = \zeta^2/C_{\text{DP}}$. For the second claim, a release $\mathbf{e}$ with total $S > 0$ has $\boldsymbol{\alpha} = \mathbf{e}/S$, so $\|\boldsymbol{\alpha} - \boldsymbol{\pi}\|_2 = \|\mathbf{e} - S\boldsymbol{\pi}\|_2/S$. Minimizing it over the closed convex set $C_S = \{\mathbf{e} : 0 \le e_i \le \hat{\epsilon}_i,\ \sum_i e_i = S\}$, which is non-empty when $S \le \sum_i \hat{\epsilon}_i$, is the Euclidean projection of $S\boldsymbol{\pi}$ onto C_S. Its KKT conditions give $e_i = \min\{\hat{\epsilon}_i, \max\{0, S\pi_i + \mu\}\}$ for the multiplier μ of the sum constraint, the form used in Eq. (34). ∎

D. Proof of Theorem 2

Theorem 2 (Existence of MFE). *Under Assumption 3, the privacy auction MFG admits at least one Mean-Field Equilibrium* (Φ^*, μ^*).

Proof: By (R5) the bid does not enter the drift, so the type process does not depend on the strategies. Its law μ_t solves the FP equation (S3) with drift $f(s,\mu_t)$, a McKean–Vlasov equation with reflection at the boundary of the compact convex set $\mathcal{S}$ (R6). Under the Lipschitz condition (R1) it has a unique solution μ^* [12]. Given μ^*, the Hamiltonian splits as $\max_b[\tilde{p}(b;\mu_t^*) - c(v,\epsilon(b))] + (\nabla_s\Phi)^\top f(s,\mu_t^*)$. The maximum is attained because $\mathcal{B}$ is compact (R2) and $\tilde{p}$ and c are continuous in b (R3), and a measurable maximizer $b^*(s,t;\mu^*)$ exists by the measurable maximum theorem [13]. Because the bid does not affect the state, the value of the representative agent is $\Phi^*(s,t) = \mathbb{E}\big[\int_t^T h(\mathbf{s}(r),r)\,dr \mid \mathbf{s}(t) = s\big]$, with $h(s,r) = \max_b[\tilde{p}(b;\mu_r^*) - c(v,\epsilon(b))]$ and $\mathbf{s}$ the reflected type process; Φ^* is the viscosity solution of the HJB equation (S2), which is linear in Φ here [14]. Thus b^* maximizes the Hamiltonian given μ^*, and μ^* solves the FP equation under b^*, so (Φ^*, μ^*) is an MFE. ∎

E. Proof of Theorem 3

Theorem 3 (MFG Approximation of Nash Equilibrium). *Let Assumptions 3 and 4 hold, and let* (Φ^*, μ^*) *be a MFE. Define the MFE-induced strategy profile* $b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*)$, $i = 1, \dots, N$. *Then* $\mathbf{b}^{N,*} = (b_1^{N,*}, \dots, b_N^{N,*})$ *is an* ε_N*-Nash Equilibrium (Definition 5) of the PAG, where*

$$\varepsilon_N = \frac{2L_u C \log N}{\sqrt{N-1}} = \mathcal{O}\left(\frac{\log N}{\sqrt{N}}\right), \tag{S5}$$

and $C > 0$ *depends only on the diameter of the compact state space* $\mathcal{S} \subset \mathbb{R}^2$.

Proof: Fix arbitrary $i \in \mathcal{N}$ and $\hat{b}_i \in \mathcal{B}$, and let $N \ge 3$.

Step 1 (Empirical measure concentration). Fix the round t and write μ^* for μ_t^*. By (R5), the states $\{\mathbf{s}_j\}_{j\ne i}$ of the round are i.i.d. from μ^*. Define $\mu_{-i}^N = \frac{1}{N-1}\sum_{j\ne i}\delta_{\mathbf{s}_j}$. Since W_1 is invariant under translating both measures, we may assume that the compact set $\mathcal{S} \subset \mathbb{R}^2$ contains the origin, so every moment of μ^* satisfies $M_q(\mu^*)^{1/q} \le \operatorname{diam}(\mathcal{S})$. Theorem 1 of [15] with dimension $d = 2$, order $p = 1$ (the critical case $p = d/2$), and moment order $q = 4$ gives, for $N-1$ samples,

$$\begin{aligned}\mathbb{E}\big[W_1(\mu_{-i}^N, \mu^*)\big] &\le C_0 \operatorname{diam}(\mathcal{S})\left(\frac{\log N}{\sqrt{N-1}} + \frac{1}{(N-1)^{3/4}}\right)\\ &\le \frac{C\log N}{\sqrt{N-1}},\end{aligned} \tag{S6}$$

where C_0 is an absolute constant and $C = 2C_0\operatorname{diam}(\mathcal{S})$, because $(N-1)^{-3/4} \le (N-1)^{-1/2} \le \log N/\sqrt{N-1}$ for $N \ge 3$.

Step 2 (Utility gap decomposition).

$$\begin{aligned}&u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*})\\ &= \underbrace{u_i(\hat{b}_i, b_{-i}^{N,*}) - u^{\text{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu^*)}_{\text{(I)}}\\ &\quad + \underbrace{u^{\text{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu^*) - u^{\text{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu^*)}_{\text{(II)}}\\ &\quad + \underbrace{u^{\text{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu^*) - u_i(b_i^{N,*}, b_{-i}^{N,*})}_{\text{(III)}}.\end{aligned} \tag{S7}$$

Step 3 (Bound on (II)). By (R5), the term $(\nabla_s\Phi)^\top f$ in Eq. (23) does not depend on the bid, so $b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*)$ maximizes $u^{\text{MFG}}(\mathbf{s}_i, \cdot\,; \mu^*)$ and (II) ≤ 0.

Step 4 (Bound on (I) and (III)). By the anonymity in Assumption 4, the N-player payoff of client i for any own bid b is $u^{\text{MFG}}(\mathbf{s}_i, b; \mu_{-i}^N)$, its mean-field payoff evaluated at the empirical law of its opponents; the cost terms cancel, and the Lipschitz bound of Assumption 4 gives $|\text{(I)}| + |\text{(III)}| \le 2L_u W_1(\mu_{-i}^N, \mu^*)$.

Step 5 (Conclusion). Taking expectations and applying Step 1:

$$\mathbb{E}\Big[u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*})\Big] \le \frac{2L_u C\log N}{\sqrt{N-1}} = \varepsilon_N. \tag{S8}$$

Since this holds for all i and all $\hat{b}_i$, $\mathbf{b}^{N,*}$ is an ε_N-Nash Equilibrium. ∎

The logarithmic factor reflects the two-dimensional state: in general the rate cannot be improved to $N^{-1/2}$, since for the uniform law on the unit square the expected matching distance between N sample points and the law is of exact order $\sqrt{\log N/N}$ [16].

F. Proof of Theorem 4

Proof: Write $s = \|F_N - F\|_\infty$, $F^{-1}(\kappa) = \inf\{x : F(x) \ge \kappa\}$, and F_N^{-1} for the empirical quantile function, so that $F_N^{-1}(k/N) = v_{(k)}$.

Step 1 (quantile perturbation). For $\kappa \in (0,1]$, $|F_N^{-1}(\kappa) - F^{-1}(\kappa)| \le s/f_{\min}$. Indeed, $F_N(x) \ge \kappa$ implies $F(x) \ge \kappa - s$, hence $F_N^{-1}(\kappa) \ge F^{-1}(\kappa - s)$; and $F(F^{-1}(\kappa + s)) = \kappa + s$ implies $F_N(F^{-1}(\kappa + s)) \ge \kappa$, hence $F_N^{-1}(\kappa) \le F^{-1}(\kappa + s)$ (arguments are clipped to $[0,1]$, where $F^{-1}(0) = 0$ and $F^{-1}(1) = \bar{v}$). Since $f \ge f_{\min}$, F^{-1} is $(1/f_{\min})$-Lipschitz on $[0,1]$.

Step 2 (acceptance test). Algorithm 1 accepts k when $c(v_{(k)}, 1/(N-k)) \le B/k$, i.e., $k\, v_{(k)} \le B(N-k)$, i.e., $\Psi_N(k/N) \le 0$ with $\Psi_N(\kappa) = \kappa(F_N^{-1}(\kappa) + B) - B$. $\Psi_N(k/N)$ is strictly increasing in k, so the accepted indices form a prefix $\{1, \dots, k_N\}$ of $\{1, \dots, N-1\}$, and $k_N \ge 1$ because $v_{(1)} \le \bar{v} \le B(N-1)$ for $N \ge 1 + \bar{v}/B$. By Step 1, $|\Psi_N(\kappa) - \Psi(\kappa)| = \kappa |F_N^{-1}(\kappa) - F^{-1}(\kappa)| \le s/f_{\min}$. Moreover, $\kappa \mapsto \kappa F^{-1}(\kappa)$ is non-decreasing, so $\Psi(\kappa') - \Psi(\kappa) \ge B(\kappa' - \kappa)$ for $\kappa' \ge \kappa$, and Ψ has the unique root κ^*.

Step 3 (selection). Upper side: if $\hat{q}_N \ge \kappa^*$, then $B(\hat{q}_N - \kappa^*) \le \Psi(\hat{q}_N) \le \Psi_N(\hat{q}_N) + s/f_{\min} \le s/f_{\min}$. Lower side: if $k_N < N-1$, rejection of $k_N + 1$ gives $\Psi_N((k_N+1)/N) > 0$, hence $\Psi((k_N+1)/N) > -s/f_{\min}$, and if $(k_N+1)/N \le \kappa^*$ then $B(\kappa^* - (k_N+1)/N) < s/f_{\min}$; if $k_N = N-1$, then $\hat{q}_N = 1 - 1/N \ge \kappa^*$ because $N \ge (1-\kappa^*)^{-1}$. In all cases $|\hat{q}_N - \kappa^*| \le s/(B f_{\min}) + 1/N$, the first inequality of Eq. (29).

Step 4 (budget). The winners are paid $p_{\text{win}} = \min\{B/k_N,\, v_{(k_N+1)}/(N - k_N)\}$, so $\sum_i p_i = \min\{B,\, k_N v_{(k_N+1)}/(N - k_N)\} \le B$. If $k_N < N-1$, rejection of $k_N + 1$ gives $(k_N+1) v_{(k_N+1)} > B(N - k_N - 1)$, hence $k_N v_{(k_N+1)}/(N - k_N) > B - (B + v_{(k_N+1)})/(N - k_N) \ge B - (B + \bar{v})/(N - k_N)$; if $k_N = N-1$ the lower bound is negative and holds trivially. This is the second inequality of Eq. (29).

Step 5 (rates). The DKW inequality with Massart's constant gives $\Pr(s > x) \le 2e^{-2Nx^2}$ [17], so $s = \mathcal{O}_P(N^{-1/2})$ and $\mathbb{E}s \le \sqrt{\pi/(2N)}$. The per-winner release is $N\epsilon_{\text{win}}^{\text{out}} = 1/(1 - \hat{q}_N)$ and the per-winner payment is $N p_{\text{win}} = \min\{B/\hat{q}_N,\, v_{(k_N+1)}/(1 - \hat{q}_N)\}$. By Steps 1 and 3, $|v_{(k_N+1)} - F^{-1}(\kappa^*)| \le \big[s\big(1 + 1/(B f_{\min})\big) + 2/N\big]/f_{\min}$, and at κ^* the two arguments of the minimum coincide because $\kappa^* F^{-1}(\kappa^*) = B(1 - \kappa^*)$. Both functions are Lipschitz near $\kappa^* \in (0,1)$, so both quantities converge at rate $\mathcal{O}_P(N^{-1/2})$, and the budget gap of Step 4 is $\mathcal{O}_P(1/N)$ because $N - k_N \ge N(1 - \kappa^*)/2$ with probability tending to one. ∎

G. Proofs for Section VII-C

Proof of Proposition 2: Fix θ, B, κ, the verified data sizes, and the reports of the other clients. The selected set $\mathcal{I}_\tau$ depends only on the data sizes and is therefore fixed; write $\mathbf{e} = \Pi_S(S\hat{\boldsymbol{\pi}})$, i.e., $e_j = \min\{\hat{\epsilon}_j, \max\{0, S\hat{\pi}_j + \mu\}\}$, where $\hat{\boldsymbol{\pi}}$ are the shares renormalized within $\mathcal{I}_\tau$, $\hat{\epsilon}_j$ is set to 0 outside it, and μ is such that the entries sum to S.

(a) Monotonicity. Each coordinate of $\mathbf{f}_i$ is non-increasing in $\hat{v}_i$ and non-decreasing in $\hat{\epsilon}_i$. The encoder has non-negative weights and non-decreasing activations, so every coordinate of $\phi_{\theta_h}(\mathbf{f}_i)$, and hence of b_{MFG}, moves in the same directions; since $\mathbf{a}(B, \kappa) \ge 0$ and σ is increasing, so does the first argument of the minimum in Eq. (33). The second argument does not depend on the reports, and the third is non-decreasing in $\hat{\epsilon}_i$. Hence S is non-increasing in $\hat{v}_i$ and non-decreasing in $\hat{\epsilon}_i$. Two properties of the projection complete the argument. (i) *Every e_j is non-decreasing in S.* On an interval of S on which the set F of entries strictly between their bounds is fixed, F is non-empty, and $\sum_{j \in F}(\hat{\pi}_j + d\mu/dS) = 1$ gives $d\mu/dS = (1 - \sum_{j \in F} \hat{\pi}_j)/|F| \ge 0$; every argument $S\hat{\pi}_j + \mu$ is therefore non-decreasing, and since μ is continuous in S, this holds globally. (ii) *At fixed S, e_i is non-decreasing in $\hat{\epsilon}_i$.* If $e_i < \hat{\epsilon}_i$, a larger cap leaves the solution unchanged. If $e_i = \hat{\epsilon}_i$ and a larger cap gave $e_i' < e_i$, then, because a larger cap can only lower μ, every other entry would not increase, and the entries would sum to less than S, a contradiction. Since $\hat{v}_i$ enters only through S, (i)–(ii) and the monotonicity of S give the stated directions for e_i, and $\epsilon_i^{\text{out}} = e_i \mathbf{1}\{e_i \ge \epsilon_{\min}\}$ is a non-decreasing function of e_i.

(b) DSIC. Fix the type $t_i = (v, \epsilon, d)$ and a report $(\hat{v}, \hat{\epsilon}) \in \mathcal{B}_i(t_i)$, and let $e(s)$ be the release at valuation report s with $\hat{\epsilon}$ and the verified size d held fixed, which is non-increasing by (a). The dividend h_i does not depend on client i's report: $\bar{S}_i$ fixes the price at 0 and the budget at $\bar{\epsilon}$, and the share depends only on the verified sizes. The utility is $u = (\hat{v} - v)e(\hat{v}) + \int_{\hat{v}}^{\bar{v}} e(s)\, ds + h_i$. If $\hat{v} \ge v$, then $(\hat{v} - v)e(\hat{v}) \le \int_v^{\hat{v}} e(s)\, ds$; if $\hat{v} < v$, then $(\hat{v} - v)e(\hat{v}) \le -\int_{\hat{v}}^{v} e(s)\, ds$. In both cases $u \le \int_v^{\bar{v}} e(s)\, ds + h_i$, which is the sufficiency direction of Proposition 1 with $h = h_i$. By (a), the release at every s is non-decreasing in the declared budget, and $\hat{\epsilon} \le \epsilon$. The bound is therefore at most the truthful utility $\int_v^{\bar{v}} \epsilon_i^{\text{out}}(s; \epsilon)\, ds + h_i$. The release already includes the effect of the report on b_{MFG}, so no further term arises.

(c) IR. The truthful utility is $\int_v^{\bar{v}} \epsilon_i^{\text{out}}(s)\, ds + h_i \ge h_i \ge 0$, where $h_i \ge 0$ because $\bar{S}_i \le B/\bar{v}$ by Eq. (33).

(d) BF. Since $\epsilon_i^{\text{out}}(s) \le \epsilon_i^{\text{out}}(\hat{v}_i)$ for $s \ge \hat{v}_i$, the first two terms of p_i are at most $\hat{v}_i \epsilon_i^{\text{out}}(\hat{v}_i) + (\bar{v} - \hat{v}_i)\epsilon_i^{\text{out}}(\hat{v}_i) = \bar{v}\epsilon_i^{\text{out}}(\hat{v}_i)$, and they sum to at most $\bar{v} \sum_i e_i = \bar{v} S$. By (a), the price 0 and the budget $\bar{\epsilon}$ give client i's largest purchase, so $\bar{S}_i \ge S$ for every i, and $\sum_i h_i \le (B - \bar{v}S) \sum_i \hat{\pi}_i = B - \bar{v}S$. Hence $\sum_i p_i \le B$. ∎

Proof of Corollary 2: Let T_n be the trapezoid estimate of $I = \int_{\hat{v}}^{\bar{v}} e(s)\, ds$ on n equal intervals of length $h = (\bar{v} - \hat{v})/n$. For a non-increasing e, T_n is the average of the left and right Riemann sums, which bracket I and differ by $h\,(e(\hat{v}) - e(\bar{v}))$, so $|T_n - I| \le h\, e(\hat{v})/2 \le \bar{v}\hat{\epsilon}/(2n)$, because the release never exceeds the declared budget. Also $0 \le T_n \le h \sum_{k<n} e(s_k) \le (\bar{v} - \hat{v})e(\hat{v})$, so (c) and (d) hold with the trapezoid payment; the dividend is computed exactly. For any report in $\mathcal{B}_i(t_i)$, the utility under the trapezoid payment differs from the exact one by at most $\bar{v}\hat{\epsilon}/(2n) \le \bar{v}\epsilon_i/(2n)$, both at that report and at the truthful one. Since truthful reporting maximizes the exact utility (Proposition 2), $\text{rgt}_i \le \bar{v}\epsilon_i/n$. ∎

Proof of Proposition 3: (a) is Markov's inequality for the non-negative variable rgt_i. For (b), fix t_i and $b' \in \mathcal{B}_i(t_i)$; then $\mathbb{E}[u_i(t_i, (b', \mathbf{t}_{-i})) - u_i(t_i, \mathbf{t}) \mid t_i] \le \mathbb{E}[\max_{b''} u_i(t_i, (b'', \mathbf{t}_{-i})) - u_i(t_i, \mathbf{t}) \mid t_i] = \mathbb{E}[\text{rgt}_i \mid t_i]$; taking the supremum over b' and the expectation over t_i gives the ex-ante statement. (c) holds because each restricted deviation set is a subset of $\mathcal{B}_i(t_i)$. For (d), let σ be any strategy that, in round τ, maps client i's current type $t_i^{(\tau)}$

and the history h_τ of past reports and outcomes to a report $b_i^{(\tau)} \in \mathcal{B}_i(t_i^{(\tau)})$. The deployed network keeps no state across rounds, so the round-τ outcome depends only on the round-τ reports, and with truthful opponents the round-τ profile $\mathbf{t}^{(\tau)}$ is independent of h_τ. The round-τ gain over truthful reporting is therefore $u_i(t_i^{(\tau)}, (b_i^{(\tau)}, \mathbf{t}_{-i}^{(\tau)})) - u_i(t_i^{(\tau)}, \mathbf{t}^{(\tau)}) \le \mathrm{rgt}_i(\theta; \mathbf{t}^{(\tau)})$ pointwise, whatever h_τ is. Summing over $\tau = 1, \dots, T$ gives the pathwise bound, and taking expectations over the i.i.d. profiles gives $T\, r_i(\theta)$. ∎

H. Proof of Proposition S1

Proof: RDP composes additively over adaptively chosen mechanisms [18], which gives ρ_i^{tot}; a round in which client i is not selected releases nothing about its data and adds nothing. An (α, ρ)-RDP mechanism is $(\rho + \log(1/\delta')/(\alpha - 1), \delta')$-DP [18], and minimizing over α gives Eq. (S1). For a Gaussian release with sensitivity Δf and $\sigma_i^{(t)} = \Delta f \sqrt{2\ln(1.25/\delta)}/\epsilon_i^{(t),\mathrm{out}}$, $\rho_i^{(t)}(\alpha) = \alpha \Delta f^2/(2(\sigma_i^{(t)})^2) = \alpha(\epsilon_i^{(t),\mathrm{out}})^2/(4\ln(1.25/\delta))$, so $\rho_i^{\mathrm{tot}}(\alpha) = \alpha\Gamma_i$ with Γ_i of Eq. (41). Writing $\alpha = 1 + x$, the bound $\Gamma_i + x\Gamma_i + \log(1/\delta')/x$ is minimized at $x = \sqrt{\log(1/\delta')/\Gamma_i}$, where it equals $\Gamma_i + 2\sqrt{\Gamma_i \log(1/\delta')}$, which is Eq. (41). ∎

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